EE582 Homework 01 Report

Aryan Ritwajeet Jha

24 September 2025

Contents

Question 1

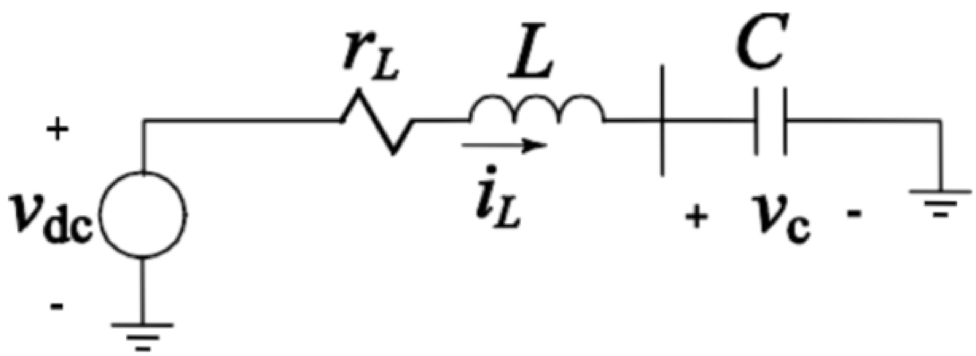

Figure 1: Circuit diagram for a simple filter for questions 1-8

A shunt harmonic filter (series RLC) is shown in Figure 1 above. It is designed for a three-phase 13.8 kV 60 Hz system to shunt the 7th harmonic current. Find the reactive power, damped and undamped resonance frequency, and quality factor of the filter. The circuit parameters are: $r_L = 1\ \Omega$, $L = 19$ mH, and $C = 8.2\ \mu$F.

Step 1: Write the KVL equation:

$$V_{\text{dc}} - r_L i_L - L\frac{di_L}{dt} - v_C = 0$$

Or equivalently:

$$L\frac{di_L}{dt} + r_L i_L + v_C = V_{\text{dc}}$$

With $i_L = C\frac{dv_C}{dt}$ and $v_C = \frac{1}{C}\int i_L dt$.

Step 2: Standard second-order ODE:

$$\frac{d^2 i_L}{dt^2} + \frac{r_L}{L}\frac{di_L}{dt} + \frac{1}{LC}i_L = 0$$

Step 3: Characteristic equation:

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$$

Step 4: Calculate parameters:

$$\omega_n = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{19\times10^{-3}\times8.2\times10^{-6}}} = 2533.5 \text{ rad/s}$$

$$\zeta = \frac{r_L}{2\sqrt{\frac{L}{C}}} = \frac{1}{2\sqrt{\frac{19\times10^{-3}}{8.2\times10^{-6}}}} = 0.0104$$

$$\omega_d = \omega_n\sqrt{1-\zeta^2} = 2533.3 \text{ rad/s}$$

Quality factor:

$$Q_{\text{factor}} = \frac{\omega_n L}{r_L} = \frac{2533.5 \times 0.019}{1} = 48.136$$

Eigenvalues (roots of characteristic equation):

$$\lambda = -\zeta\omega_n \pm j\omega_d = -26.316 \pm 2533.3j$$

Reactive Power Computation: The reactive power is computed for a three-phase system. First, the per-phase voltage $V_{ph} = V_{LL}/\sqrt{3}$ and current I_{ph} are calculated using the per-phase impedance $Z_{ph} = r_L + j(X_L - X_c)$. Then, the reactive power is determined as:

$$Q_{\text{kVAr}} = 3 \times |I_{ph}|^2 \times X_c$$

where $X_c = \frac{1}{2\pi f C}$ and $X_L = 2\pi f L$ at $f = 60$ Hz.

$$Q_{\text{kVAr}} = -602.04 \text{ kVAr}$$

Negative sign indicates the filter is absorbing reactive power (capacitive).
All calculations confirm the system is very underdamped ($\zeta = 0.0104$).

Summary of Key Results:

- Natural frequency: $\boxed{\omega_n = 2533.5 \text{ rad/s}}$
- Damping ratio: $\boxed{\zeta = 0.0104}$
- Damped frequency: $\boxed{\omega_d = 2533.3 \text{ rad/s}}$
- Quality factor: $\boxed{Q = 48.136}$
- Eigenvalues: $\boxed{-26.316 \pm 2533.3j}$
- Reactive power: $\boxed{-602.04 \text{ kVAr}}$

Question 2

Calculate the transient response of the circuit shown in Fig. 1(a) to a step voltage change if $v_{dc}(t) = s(t) = 11.3$ kV for $t \geq 0$. Comment on the eigenvalues and response type.

Referring again to the series RLC circuit in Q1 (Figure 1) and writing relevant KVL and current equations, we have:

KVL:

$$V_{\mathrm{dc}} - r_L i_L - L\frac{di_L}{dt} - v_C = 0 \tag{1}$$

where i_L is the inductor current and v_C is the capacitor voltage.

Capacitor current:

$$i_L = C\frac{dv_C}{dt} \tag{2}$$

Substitute (2) into (1):

$$V_{\mathrm{dc}} - r_L C\frac{dv_C}{dt} - LC\frac{d^2 v_C}{dt^2} - v_C = 0$$

Rearrange to standard form:

$$LC\frac{d^2 v_C}{dt^2} + r_L C\frac{dv_C}{dt} + v_C = V_{\mathrm{dc}}$$

Divide both sides by LC:

$$\frac{d^2 v_C}{dt^2} + \frac{r_L}{L}\frac{dv_C}{dt} + \frac{1}{LC}v_C = \frac{V_{\mathrm{dc}}}{LC}$$

This is a standard second-order nonhomogeneous ODE. The homogeneous solution is:

$$\frac{d^2 v_C}{dt^2} + \frac{r_L}{L}\frac{dv_C}{dt} + \frac{1}{LC}v_C = 0$$

Eigenvalue computation: The characteristic equation is:

$$s^2 + \frac{r_L}{L}s + \frac{1}{LC} = 0$$

The eigenvalues are:

$$s = \frac{-\frac{r_L}{L} \pm \sqrt{\left(\frac{r_L}{L}\right)^2 - 4 \cdot \frac{1}{LC}}}{2}$$

For the given values ($r_L = 1\ \Omega$, $L = 19$ mH, $C = 8.2\ \mu$F):

$$\frac{r_L}{L} = \frac{1}{0.019} = 52.63\ \mathrm{s}^{-1}$$
$$\frac{1}{LC} = \frac{1}{0.019 \times 8.2 \times 10^{-6}} = 6,430,041\ \mathrm{s}^{-2}$$

Plug into the quadratic formula:

$$s = \frac{-52.63 \pm \sqrt{(52.63)^2 - 4 \times 6,430,041}}{2}$$

The discriminant is negative, so the roots are complex conjugates:

$$s = -26.316 \pm j\,2533.3$$

Interpretation: The eigenvalues are complex conjugates with negative real parts, confirming the circuit is underdamped. Based on the very low value of the damping ratio $\zeta = 0.0104$ calculated in Q1, the transient response will exhibit oscillatory behavior with a slow decay.

General solution:

$$v_C(t) = v_{C,\infty} + [v_{C,0} - v_{C,\infty}] \left[1 - \frac{1}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} \sin(\omega_d t + \phi)\right]$$

where $v_{C,\infty}$ is the steady-state value, $v_{C,0}$ is the initial value, and $\omega_d = \omega_n\sqrt{1-\zeta^2}$.

For a step input, $v_{C,\infty} = x_{SS}$, and the solution simplifies to:

$$\boxed{x(t) = x_{SS} \left[1 - \frac{1}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} \sin(\omega_d t + \phi)\right]}$$

For the given values ($r_L = 1\ \Omega$, $L = 19$ mH, $C = 8.2\ \mu$F), the numeric solution is:

$$\boxed{x(t)[kV] = 11.3 * \left[1 - 1.00005\, e^{-26.34\,t}\ \sin(2533.3\,t + 1.5604)\right]}$$

Question 3

Develop a Simulink model of the circuit shown in Fig. 1(a) using conventional Simulink blocks (i.e., summers, gains, integrals, etc.). Compute the transient response (solutions) on time interval $[0, 0.1]$ s using the ode3 solver with 0.1 ms time-step size.

Block Diagram

The following block diagram represents the Simulink implementation of the RLC circuit for part (a):

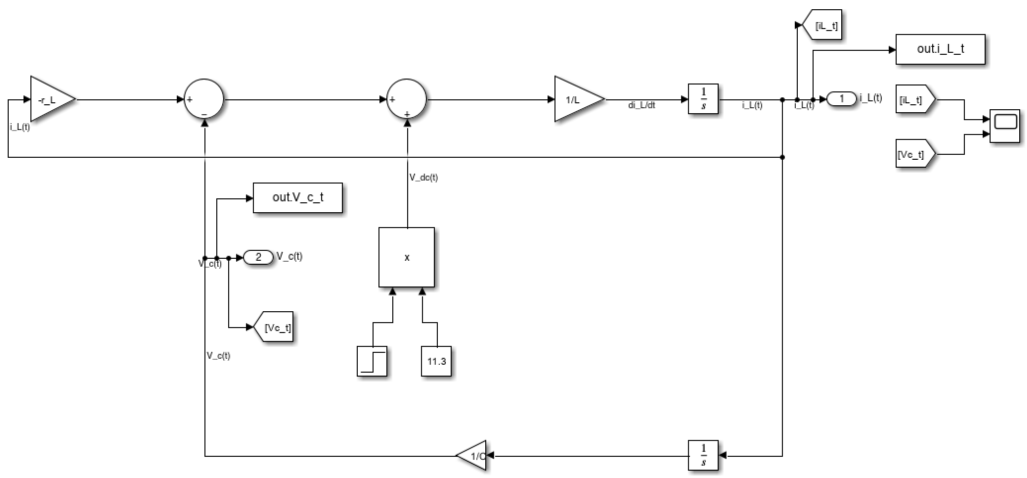

Figure 2: Simulink block diagram for the series RLC circuit (part a).

Simulation Results

The transient response of the circuit was simulated in Simulink using the `ode3` solver with a fixed time step of 0.1 ms over the interval $[0, 0.1]$ s. The following plot shows the inductor current $i_L(t)$ and capacitor voltage $v_C(t)$:

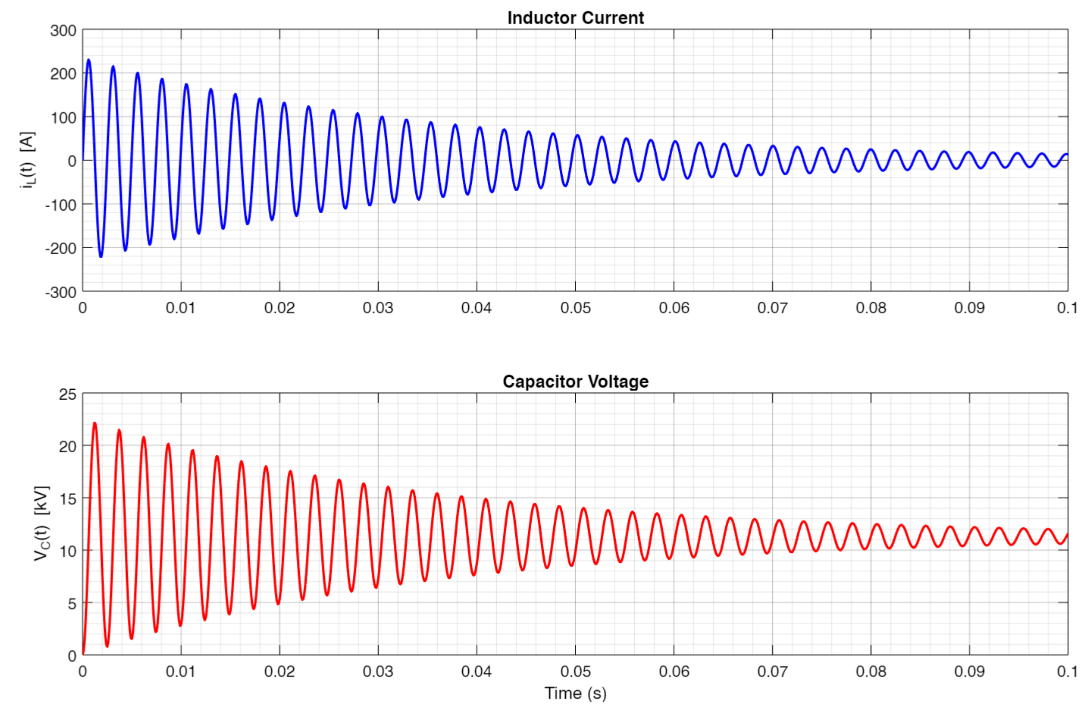

Figure 3: Simulink simulation results: $i_L(t)$ and $v_C(t)$ for the RLC circuit.

Discussion

The simulation results show the expected underdamped oscillatory response, consistent with the analytical solution derived in Q1 and Q2. The current and voltage waveforms exhibit oscillations with a slow decay, as predicted by the low damping ratio ($\zeta = 0.0104$). The Simulink model accurately captures the circuit dynamics using only basic blocks and the specified solver settings.

Question 4

Find the system eigenvalues using the MATLAB functions `linmod` and `eig` on the command line. Compare the results with part 2.

MATLAB Output

The following figure shows the eigenvalues computed using MATLAB's `linmod` and `eig` functions:

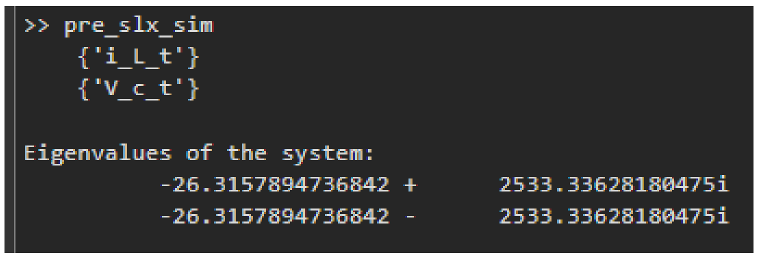

Figure 4: Eigenvalues of the system computed using `linmod` and `eig` in MATLAB.

Comparison with Analytical Results

The eigenvalues obtained from MATLAB are:

$$-26.3158 \pm 2533.34i$$

These match very closely with the hand-computed values in Q2:

$$-26.316 \pm 2533.3i$$

The small numerical difference is due to rounding and floating-point precision in MATLAB. Both methods confirm the system is underdamped with complex conjugate eigenvalues, as expected from the analytical derivation.

Question 5

Analyze the stability of the Forward Euler (ode1) and explicit RK5 (ode5) methods for the RLC circuit. Plot the stability regions and indicate the range of h for which the methods are stable.

Stability Plots

The following figure shows the stability regions for Forward Euler (ode1) and explicit RK5 (ode5) methods, with the scaled eigenvalues $z = h\lambda$ plotted for various step sizes h:

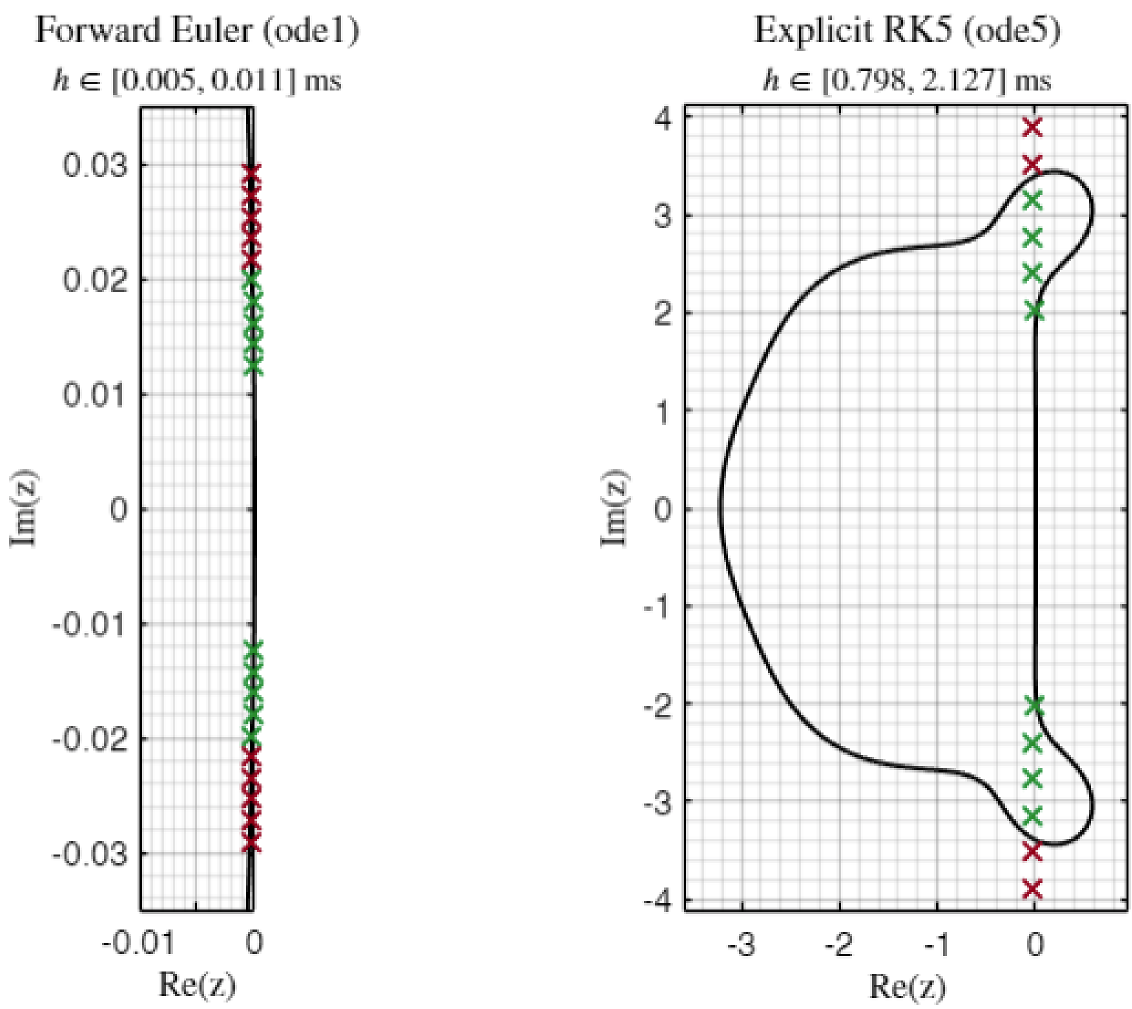

Figure 5: Stability regions for Forward Euler (ode1, left) and explicit RK5 (ode5, right). Each cross represents $z = h\lambda$ for a candidate step size h and the system eigenvalues λ. Green crosses are inside the stability region (stable), red crosses are outside (unstable). Note that the plot for Forward Euler is zoomed in to show the small stability region.

Stability Criteria and z Formulae

For a linear system $\dot{x} = \lambda x$, the stability of a numerical method is determined by the location of $z = h\lambda$ in the method's stability region:

- **Forward Euler (FE):** The stability function is $R(z) = 1 + z$. The method is stable if $|1 + z| < 1$.
- **Explicit RK5 (Taylor-5):** The stability function is $R(z) = 1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \frac{z^4}{4!} + \frac{z^5}{5!}$. The method is stable if $|R(z)| < 1$. (Note: MATLAB's ode5 is not exactly a Taylor-5 method, but the Taylor-5 region is a good approximation for illustration.)

Interpretation of Crosses

Each cross in the plot corresponds to a value of $z = h\lambda$ for a particular step size h and one of the system's eigenvalues λ.
Green crosses indicate z values inside the stability region (method is stable for that h), while **red crosses** are outside (unstable for that h).

Maximum Stable Step Size

From the plot and MATLAB output:

- **Euler:** $h_{\max} \approx 0.008$ ms
- **Taylor-5:** $h_{\max} \approx 1.329$ ms

These values indicate the largest time step for which each method remains stable for the given system.

Summary

The explicit RK5 (Taylor-5) method allows for a much larger stable time step than Forward Euler, as seen from the much larger stability region. The actual ode5 implementation in MATLAB is not exactly Taylor-5, but the qualitative stability behavior is similar for this analysis.

Question 6

Use any one of the fixed-time-step solvers of your own choice. Plot the solution for i_L and v_C. Experimentally establish the largest time step $h_{\max}$ that produces a stable and convergent solution that you can trust. Compare your result with the result of using the ode5 at $h = 0.5 \times 10^{-3}$ s and $h = 1.0 \times 10^{-3}$ s.

Simulation Results

The following figures show the results of the fixed-step simulations:

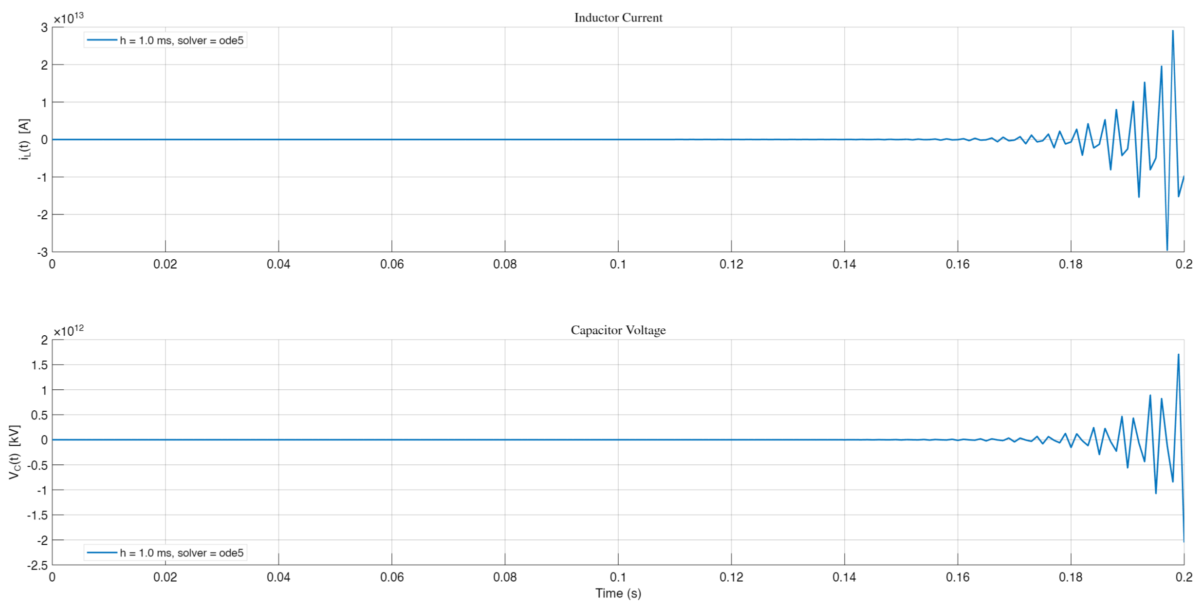

Figure 6: Diverging solution for $h = 1$ ms using ode5 (RK5).

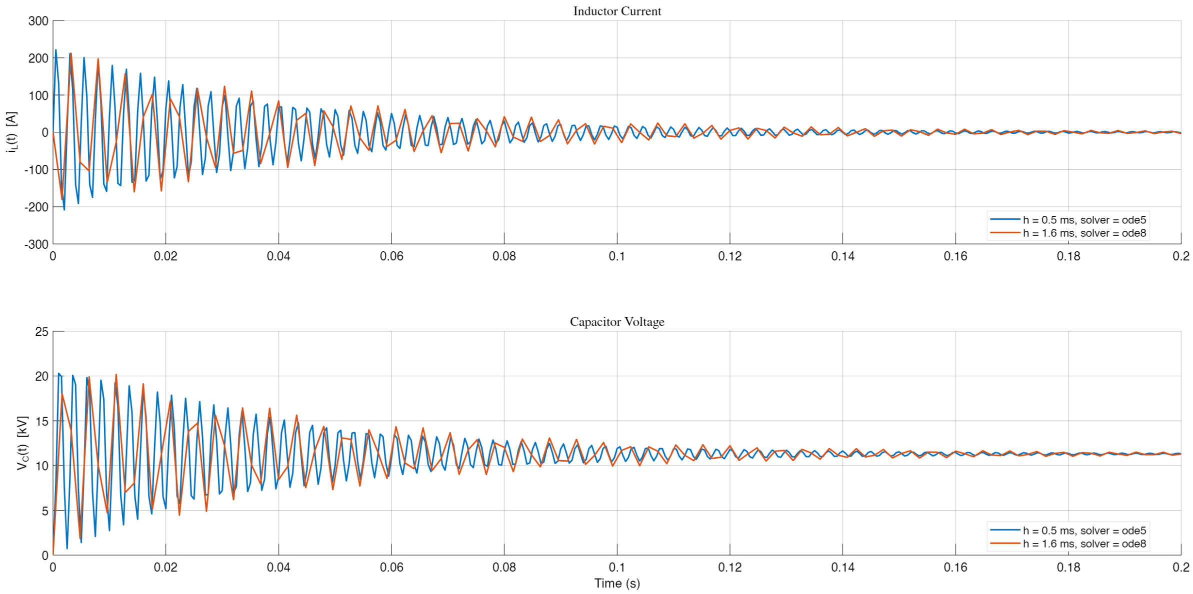

Figure 7: Converging solutions for RK8 with $h = 1.6$ ms and ode5 with $h = 0.5$ ms. The legend indicates the solver and step size.

Discussion

For this part, I used an explicit RK8 method for my own solver. The maximum time step $h_{\max}$ for which the RK8 method produced a stable and accurate solution was determined experimentally to be 1.6 ms. For ode5 (RK5), the $h = 1$ ms case is shown separately above, as it diverges and would spoil the scale of the converging plot.

The experimentally obtained $h_{\max}$ for RK8 was found to be less than the theoretical value for Taylor-5 (≈ 1.3 ms). For ode5 (RK5), the maximum stable h was found to be less than 0.7 ms (figure not shown here), which is also less than the theoretical prediction. The theoretical values are based on the Taylor-5 stability region, but ode5 is not exactly Taylor-5, and RK8 has a different (larger) stability region.

Remark

The fact that the experimentally determined $h_{\max}$ is smaller than the theoretical value is mainly due to (i) the difference between the actual Dormand–Prince implementation of `ode5` and the idealized Taylor-5 surrogate (their stability regions are not identical), and (ii) the oscillatory eigenvalues of the system, where reliable simulation typically requires ∼20–30 steps per cycle. For our case, $f_d \approx 403$ Hz so $T = 1/f_d \approx 2.48$ ms; thus an accuracy-driven step is

$$h \lesssim \frac{T}{20\text{–}30} \;\approx\; 0.08\text{–}0.12 \text{ ms}.$$

By contrast, a theoretical $h_{\max}$ inferred from the Taylor-5 absolute-stability boundary is much larger (on the order of ∼0.5–1.0 ms in our plots), which explains the discrepancy with the empirically safe step used in simulation.

Question 7

Using a variable-time-step solver of your choice, establish a convergent solution that you can trust. Plot the solution i_L and v_C, and the time step h as functions of time. Your results should be accurate enough to be assumed as reference solution.

Discussion

Since I initially don't have a notion of what a 'good' solver output looks like for this problem, the only 'reference' transient response I'm assuming is the one I can obtain by plotting the transient function derived in Q2 – I use that as a base for comparison in Q8.

After experimenting with all variable step solvers in MATLAB (ode45, ode23, ode113, ode15s, ode23s, ode23t, ode23tb), my findings are somewhat inconclusive. I personally liked stiff variations of ode23 (ode23s, ode23t, ode23tb). All subsequent comparison and discussion, including the reasoning behind my choice is in Q8.

Question 8

Using variable-step solvers, set $h_{max} = 1 \times 10^{-2}$ and both error tolerances $\varepsilon_{abs} = \varepsilon_{rel} = 10^{-3}$. Experiment with all the available variable-step solvers. In a table, print the maximum time-step size h_{max} chosen by each solver and the total number of steps taken. Comment on how these solutions compare with the reference solution found in part 7. Which solver in your opinion is more efficient and does the best job for this system? Compare the step-size and the overall number of steps observed in parts 6, 7, and this part.

Figures and Analysis

The following figures illustrate the transient response of the inductor current $i_L(t)$ and capacitor voltage $V_C(t)$ for both nonstiff and stiff solvers. Figure 8 includes all solvers, highlighting the performance of solvers that fail to dampen oscillations effectively. Figure 9 excludes these solvers, focusing only on the solvers that converge to the expected damped sinusoidal response:

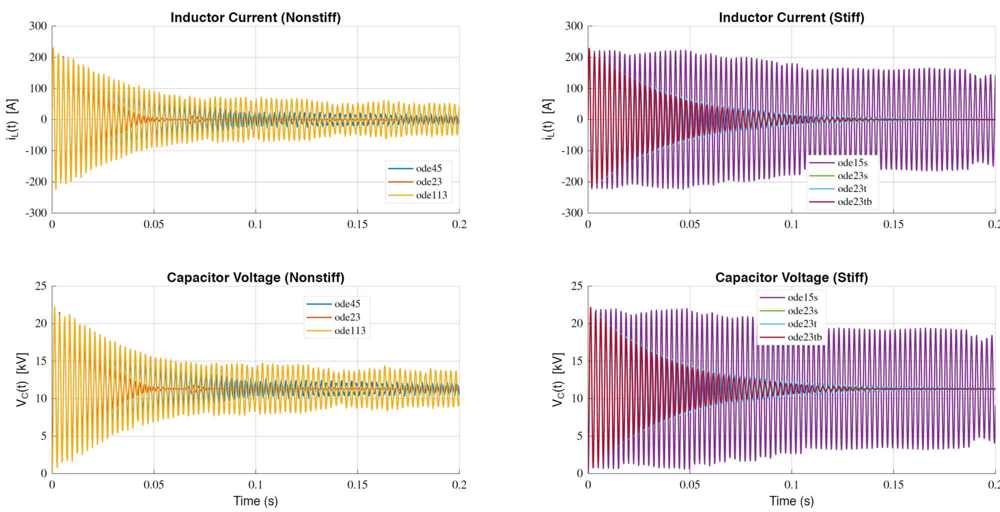

Figure 8: Transient Response of Inductor Current and Capacitor Voltage (Including Non-Dampening Solvers)

Table of Results

The table below, supported by fig. 10, summarizes the performance of each solver, including the maximum step size used, the number of steps taken, the simulation time, and the RMSE error:

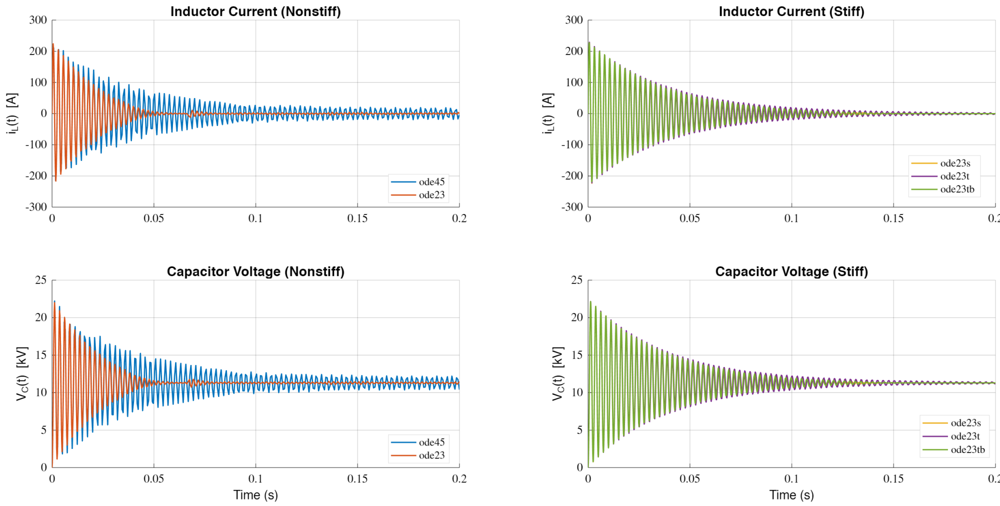

Figure 9: Transient Response of Inductor Current and Capacitor Voltage (Excluding Non-Dampening Solvers)

Solver	Type	MaxStepUsed	NumSteps	SimTime_sec	Error_kV
{'ode45' }	"Nonstiff"	0.00085483	318	7.7641	0.61709
{'ode23' }	"Nonstiff"	0.0013118	415	0.4192	1.2333
{'ode113' }	"Nonstiff"	0.00061603	755	0.37917	1.5953
{'ode15s' }	"Stiff"	0.00031737	1174	0.2778	5.3695
{'ode23s' }	"Stiff"	0.01	468	0.49326	1.9883
{'ode23t' }	"Stiff"	0.00072387	784	0.23214	2.1127
{'ode23tb'}	"Stiff"	0.01	535	0.2111	1.7507

Figure 10: Simulation Results for Variable-Step and Fixed-Step ODE Solvers

Solver	**Type**	h_{Max}	**Num Steps**	**Time [s]**	**Error [kV]**
ode45	Nonstiff	0.0008548	318	0.2753	0.6171
ode23	Nonstiff	0.0013118	415	0.3346	1.2333
ode113	Nonstiff	0.0006160	755	0.2697	1.5953
ode15s	Stiff	0.0003174	1174	0.3068	5.3695
ode23s	Stiff	0.01	468	0.3072	1.9883
ode23t	Stiff	0.0007239	784	0.2488	2.1127
ode23tb	Stiff	0.01	535	0.2490	1.7507
ode5	Fixed-step	0.0005	401	0.2556	0.2283
ode8	Fixed-step	0.0005	401	0.2388	0.0000422

Table 1: Summary of Solver Performance Including Fixed-Step RK8

Discussion

Nonstiff vs Stiff Solvers

While nonstiff solvers (ode45, ode23) generally showed lesser error, their transient response 'shape' was less consistent with a damped sinusoid as expected from the analytical solution. Stiff solvers (ode15s, ode23s) produced responses that visually aligned better with the expected damped sinusoidal shape, but they exhibited higher numerical error.

Maximum step size not necessarily proportional to error

Interestingly, nonstiff solver ode113 and stiff solver ode15s took very small steps yet resulted in a significantly higher error compared to other solvers. On the other hand, stiff solvers like ode23s and ode23tb took the largest possible steps (perhaps only occasionally or regularly), yet managed to achieve lower errors. This behavior warrants a deeper analysis of how these solvers handle step size and error accumulation.

Fixed-Step Solvers

Fixed-step solvers RK5 and RK8 demonstrated exceptional performance, with RK8 achieving the lowest error (0.0000422 kV). However, the fixed step size of 0.5 ms is overly generous compared to the variable-step solvers, making the comparison somewhat unfair. A more balanced comparison would involve reducing the fixed step size to align with the typical step sizes observed in variable-step solvers. As previously observed, a step size of $\geq$ 0.7 ms is infeasible for RK5, so a variable step size solver taking a maximum step size of 10 ms and still not diverging is in a way already superior.

Simulation Time Analysis

Moreover, the simulation times across all solvers were found to vary significantly between runs, making solve time an inconclusive metric for this system. The choice of solver should therefore depend more on the error metric and the qualitative shape of the transient response rather than computational efficiency. Among the variable step size solvers, stiff solvers like ode23s, ode23t and ode23tb remain reliable for capturing the expected response shape.

Remarks on Error Computation

The error metric used in this analysis is the root mean square error (RMSE), computed by evaluating the analytical solution at the same time steps as those of the numerical solver. The analytical solution, derived in Q2, is given by:

$$V_c(t)[kV] = 11.3\left[1 - 1.00005\, e^{-26.34\, t}\, \sin(2533.3\, t + 1.5604)\right]$$

This approach ensures a fair comparison, as the error is calculated point-by-point at identical time instances.

Conclusion - Recommended Solver

In conclusion, the stiff solvers, particularly the ode23 variants, are recommended for this problem due to their ability to handle rapid dynamics efficiently while maintaining accuracy.

Question 9

In Fig. 1(a) the equivalent series resistance and inductance of the capacitor (r_{L2} and L_2) were added to the inductance parameters r_{L1} and L_1 (i.e. $r_L = r_{L1} + r_{L2}$ and $L = L_1 + L_2$). In Fig. 1(b), they are separated and also the capacitor leakage resistance (r_c) is considered. In addition, the discharge resistance (r_{leak}) is shown in Fig. 1(b). This resistance is used to discharge the filter capacitor when it is off-line for safety purposes which was ignored in the simple model shown in Fig. 1(a). Implement the model of Fig. 1(b) in Simulink using basic blocks considering $r_{L2} = 323$ mΩ, $L_2 = 1$ mH, $r_c = 1$ MΩ, and $r_{leak} = 30$ kΩ. If the filter is disconnected, how long does it take for the capacitor to fully discharge?

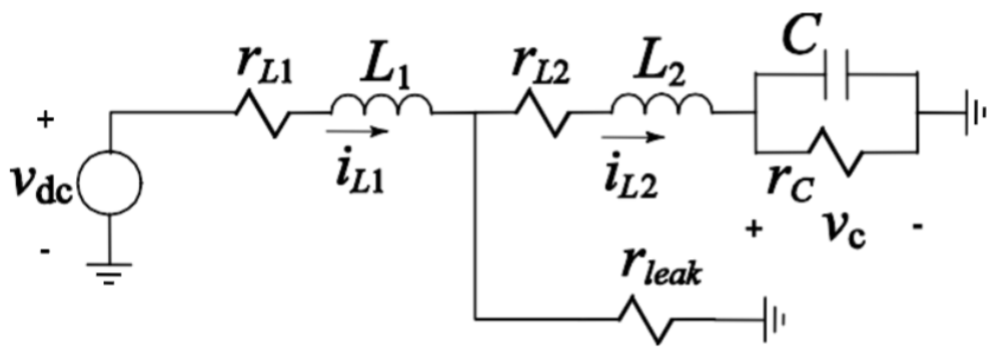

Figure 11: Circuit diagram of the full filter for questions 9-12

Figure 12 shows the Simulink block diagram for the full filter model in Figure 11. Actually this block diagram was made after hand deriving the equations for the full filter model in q10 and obtaining the state space model. This simulink model will be primarily used q10 onwards. Current question is mainly focused on the disconnected variation of this filter, which is analyzed next.

When the filter is disconnected as shown in Figure 13, the capacitor C discharges through its leakage resistance r_C and the discharge resistor r_{leak}. The relevant equations can be derived as follows.

KCL at the capacitor node:

$$i_{L2} = C\frac{dv_C}{dt} + \frac{v_C}{r_C} \tag{1}$$

KVL across L_2 and resistances:

$$-(r_{\text{leak}} + r_{L2})i_{L2} - L_2\frac{di_{L2}}{dt} = v_C \tag{2}$$

Substitute (1) into (2):

$$-(r_{\text{leak}} + r_{L2})\left(C\frac{dv_C}{dt} + \frac{v_C}{r_C}\right) - L_2\left(C\frac{d^2v_C}{dt^2} + \frac{1}{r_C}\frac{dv_C}{dt}\right) = v_C$$

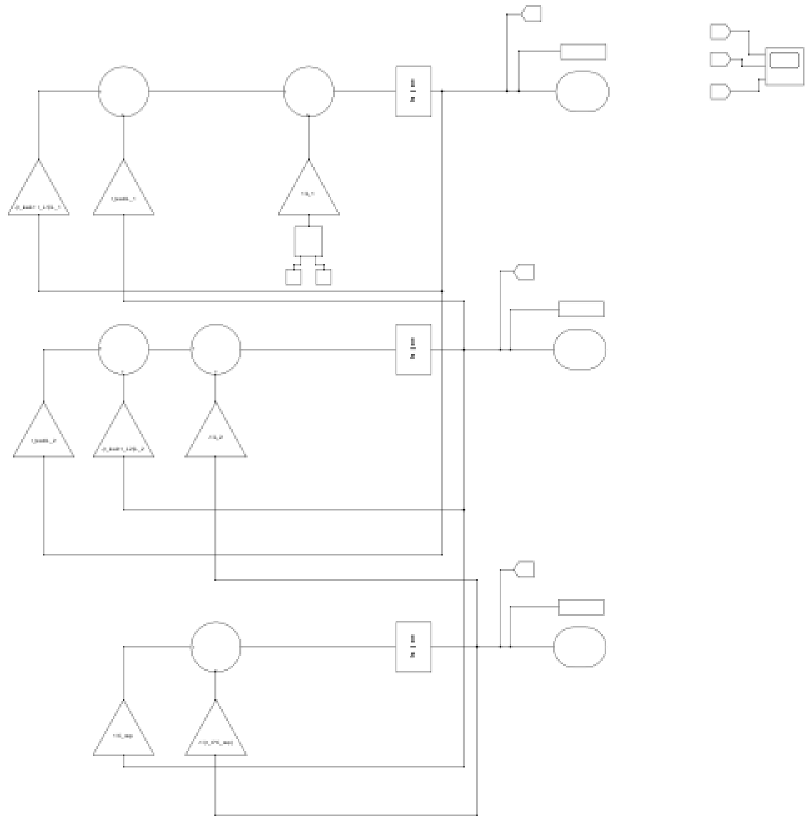

Figure 12: Simulink Block Diagram for Filter Analysis

Rearrange terms:

$$-L_2 C\frac{d^2 v_C}{dt^2} - \left[(r_{\text{leak}} + r_{L2})C + \frac{L_2}{r_C}\right]\frac{dv_C}{dt} - \left(\frac{r_{\text{leak}} + r_{L2}}{r_C} + 1\right)v_C = 0$$

Multiply through by -1:

$$L_2 C\frac{d^2 v_C}{dt^2} + \left[(r_{\text{leak}} + r_{L2})C + \frac{L_2}{r_C}\right]\frac{dv_C}{dt} + \left(\frac{r_{\text{leak}} + r_{L2}}{r_C} + 1\right)v_C = 0$$

Divide through by $L_2 C$:

$$\frac{d^2 v_C}{dt^2} + \left(\frac{r_{\text{leak}} + r_{L2}}{L_2} + \frac{1}{r_C C}\right)\frac{dv_C}{dt} + \frac{1}{L_2 C}\left(\frac{r_{\text{leak}} + r_{L2}}{r_C} + 1\right)v_C = 0$$

Final second-order ODE (disconnected case):

$$\boxed{\frac{d^2 v_C}{dt^2} + \left(\frac{r_{\text{leak}} + r_{L2}}{L_2} + \frac{1}{r_C C}\right)\frac{dv_C}{dt} + \frac{1}{L_2 C}\left(\frac{r_{\text{leak}} + r_{L2}}{r_C} + 1\right)v_C = 0}$$

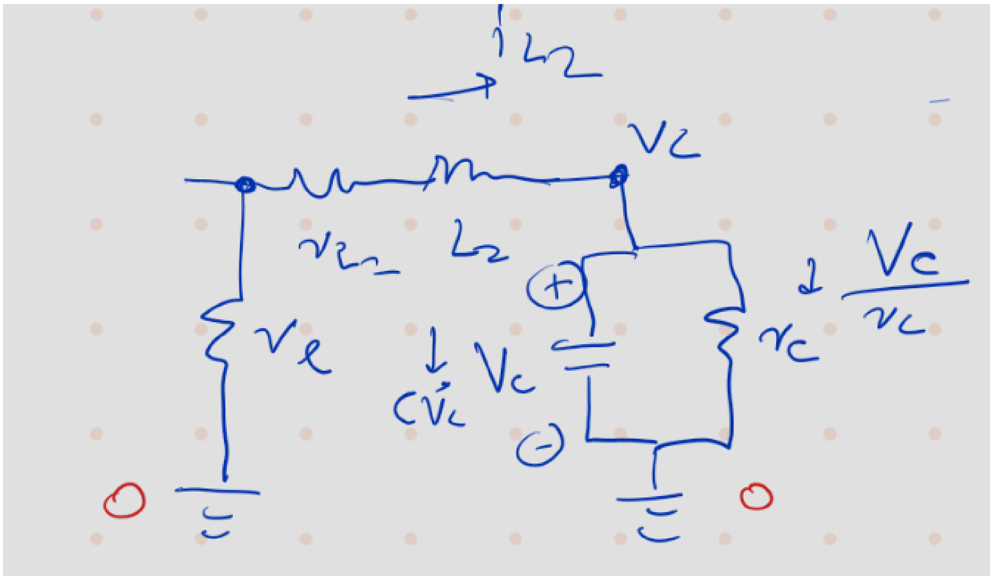

Figure 13: Hand-drawn Circuit Diagram for Disconnected Filter Analysis

This is a homogeneous second-order differential equation governing the capacitor voltage $v_C(t)$ during discharge when the filter is disconnected. The eigenvalues can be obtained by solving the characteristic quadratic:

$$s^2 + \alpha s + \beta = 0,$$

where

$$\alpha = \frac{r_{\text{leak}} + r_{L2}}{L_2} + \frac{1}{r_C C}, \quad \beta = \frac{1}{L_2 C}\left(\frac{r_{\text{leak}} + r_{L2}}{r_C} + 1\right).$$

Disconnected Filter Analysis - Overdamped Case

The disconnected filter is governed by the second-order differential equation derived earlier. Solving the characteristic equation:

$$s^2 + \alpha s + \beta = 0,$$

where

$$\alpha = \frac{r_{\text{leak}} + r_{L2}}{L_2} + \frac{1}{r_C C},$$
$$\beta = \frac{1}{L_2 C}\left(\frac{r_{\text{leak}} + r_{L2}}{r_C} + 1\right),$$

we obtain the poles:

$$s_1 = -30000318.9 \text{ s}^{-1}, \quad s_2 = -4.18694867 \text{ s}^{-1}.$$

The time constants associated with these poles are:

$$\tau_{\text{fast}} = \frac{1}{|s_1|} = 3.333 \times 10^{-8} \text{ s } (0.000 \text{ ms}),$$
$$\tau_{\text{slow}} = \frac{1}{|s_2|} = 0.238837 \text{ s } (238.837 \text{ ms}).$$

Since the system is stiff, the fast-decaying pole s_1 can be ignored for practical purposes. The discharge time is determined by the slower time constant τ_{slow}. For approximately 99.3% discharge, the total time is:

$$\tau_{\text{discharge}} = 5 \times \tau_{\text{slow}} = \boxed{1.194 \text{ s } (1194 \text{ ms})}.$$

Remark: The large damping ratio ($\zeta = 1338.39$) confirms the overdamped nature of the system, making the slower pole dominant for discharge calculations.

Question 10

Calculate the eigenvalues of the new system by hand and using MATLAB functions and repeat Part 5 for the new model of Fig. 1(b). Is this system numerically stiff? Using any fixed–time–step solver of your choice simulate the model and plot the solution for i_L and v_C. Experimentally establish the largest time step $h_{\max}$ that produces a stable and convergent solution that you can trust.

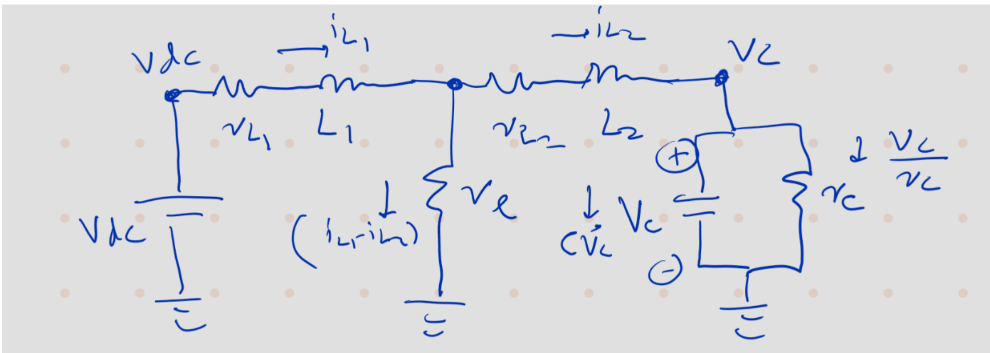

Figure 14: Hand-drawn Circuit Diagram for Filter Analysis

Referring to Figure 14, choose the states

$$x \triangleq \begin{bmatrix} i_{L1} & i_{L2} & v_C \end{bmatrix}^\top, \qquad u \triangleq v_{dc}.$$

Node voltage at the junction between L_1 and L_2 – Let's call the node $n1$ (not shown in diagram)

By KCL through the shunt r_{leak},

$$i_{r_\ell} = \frac{v_{n1}}{r_{\text{leak}}} = i_{L1} - i_{L2} \tag{1}$$

$$v_{n1} = r_{\text{leak}}\,(i_{L1} - i_{L2}). \tag{2}$$

KVL from the source to node $n1$:

$$u - r_{L1} i_{L1} - L_1 \frac{di_{L1}}{dt} - v_{n1} = 0. \tag{3}$$

KVL from node $n1$ to the capacitor node:

$$v_{n1} - r_{L2} i_{L2} - L_2 \frac{di_{L2}}{dt} - v_C = 0. \tag{4}$$

KCL at the capacitor node:

$$i_{L2} = C\,\frac{dv_C}{dt} + \frac{v_C}{r_C}. \tag{5}$$

Solve for the state derivatives. Using 3 in 4:

$$L_1 \frac{di_{L1}}{dt} = u - r_{L1} i_{L1} - r_{\text{leak}}(i_{L1} - i_{L2})$$
$$\Rightarrow \frac{di_{L1}}{dt} = \frac{1}{L_1}\left(u - (r_{L1} + r_{\text{leak}}) i_{L1} + r_{\text{leak}} i_{L2}\right). \tag{6}$$

Using 6 in 5:

$$L_2 \frac{di_{L2}}{dt} = r_{\text{leak}}(i_{L1} - i_{L2}) - r_{L2} i_{L2} - v_C$$
$$\Rightarrow \frac{di_{L2}}{dt} = \frac{1}{L_2}\left(r_{\text{leak}} i_{L1} - (r_{\text{leak}} + r_{L2}) i_{L2} - v_C\right). \tag{7}$$

Rewriting 5:

$$\frac{dv_C}{dt} = \frac{1}{C} i_{L2} - \frac{1}{r_C C} v_C. \tag{8}$$

State–space form. Collecting 6–8 in $\dot{x} = Ax + Bu$ gives

$$\boxed{A = \begin{bmatrix} -\frac{r_{L1} + r_{\text{leak}}}{L_1} & \frac{r_{\text{leak}}}{L_1} & 0 \\ \frac{r_{\text{leak}}}{L_2} & -\frac{r_{\text{leak}} + r_{L2}}{L_2} & -\frac{1}{L_2} \\ 0 & \frac{1}{C} & -\frac{1}{r_C C} \end{bmatrix}, \qquad B = \begin{bmatrix} \frac{1}{L_1} \\ 0 \\ 0 \end{bmatrix}.}$$

Eigenvalues of the full model (Figure 14).

For the given numerical values ($r_L = 1\ \Omega$, $r_{L2} = 0.323\ \Omega$, $L = 19$ mH, $L_2 = 1$ mH, $C = 8.2\ \mu$F, $r_{\text{leak}} = 30$ kΩ, $r_C = 1$ MΩ) we obtain:

$$\boxed{\lambda(A_{\text{hand}}) = \left\{ \begin{array}{l} -\,3.16669709979566 \times 10^7, \\ -\,28.200886223932 \pm j\,2533.33367805764 \end{array} \right\} \text{ s}^{-1}.}$$

Linearizing the Simulink model with `linmod` and computing `eig` returns the *same* set of eigenvalues:

$$\boxed{\lambda(A_{\text{lin}}) = \left\{-3.16669709979566 \times 10^7,\ -28.200886223932 \pm j\,2533.33367805764\right\} \text{ s}^{-1}.}$$

Comment on numerical stiffness. The system is strongly stiff: the ratio between the fastest and the slow time scales is

$$\kappa \equiv \frac{|\Re\{\lambda_{\text{fast}}\}|}{|\Re\{\lambda_{\text{slow}}\}|} \approx \frac{3.17 \times 10^7}{28.2} \approx 1.1 \times 10^6.$$

One real pole at -3.17×10^7 s^{-1} (time constant $\tau_{\text{fast}} \approx 31.6$ ns) coexists with a lightly–damped oscillatory pair ($\omega_d \approx 2533$ rad/s). This wide separation forces very small steps for explicit fixed–step solvers.

Fixed–step simulation. Using a conservative explicit step such as $h = 10^{-4}$ s (0.1 ms) with a 5th–order explicit method produced stable, convergent waveforms for i_{L1}, i_{L2} and v_C over 0.2 s (see Fig. 15). Larger h begins to corrupt the phase and envelope due to the lightly–damped oscillatory pair.

Using the implicit solver `ode14x`, a maximum step size of $h = 10^{-4}$ s (0.1 ms) produced stable, convergent waveforms for i_{L1}, i_{L2}, and v_C over 0.2 s (see Fig. 15). However, the quality of the solution begins to degrade at this upper limit of h. A smaller step size of $h = 10^{-5}$ s (not shown here) yielded significantly better results, with improved accuracy and stability in the waveforms.

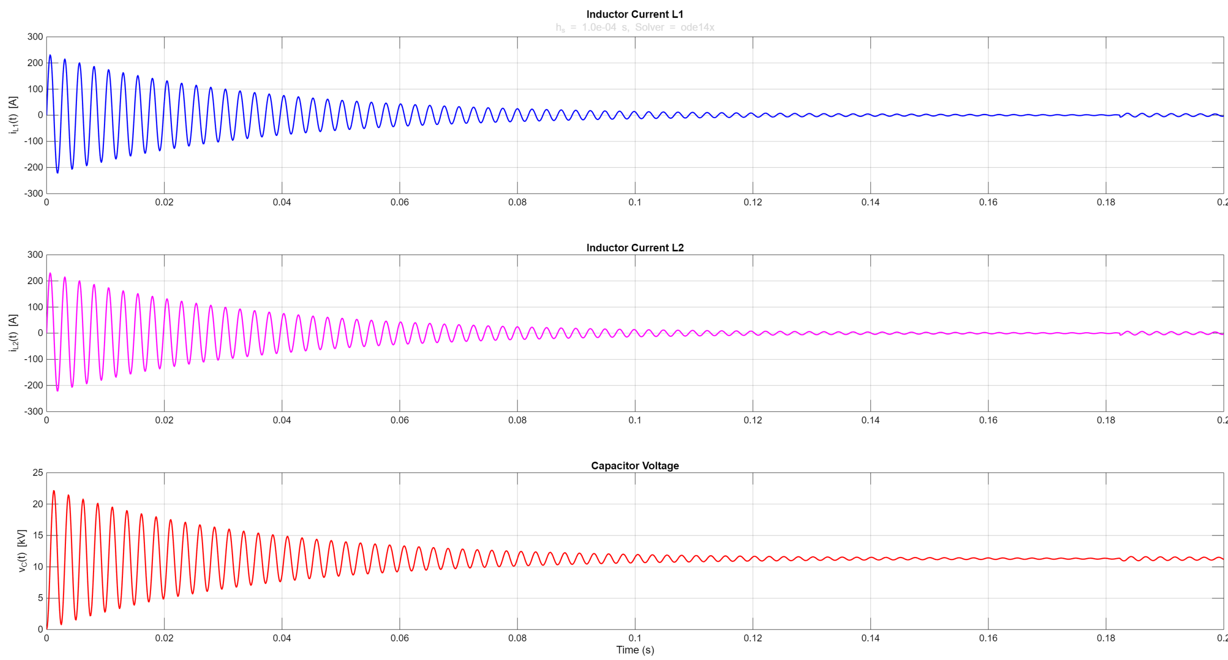

Figure 15: Fixed–step simulation of the full model (Figure 11); solver = ode14x, $h = 10^{-4}$ s, $T_{\text{horizon}} = 0.2$ s. Excuse the really light subtitle text.

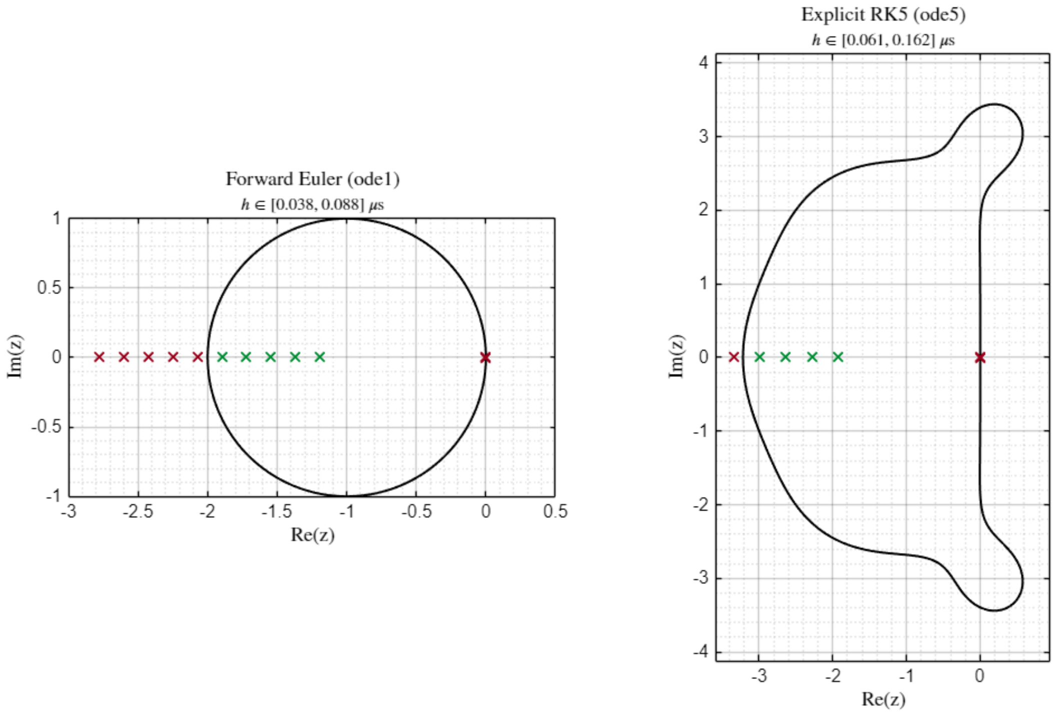

Figure 16: Stability regions for Forward Euler (ode1, left) and explicit RK5 (ode5, right). Each cross represents $z = h\lambda$ for a candidate step size h and the system eigenvalues λ. Green crosses are inside the stability region (stable), red crosses are outside (unstable). Note that this time due to the abnormally large eigenvalue, the plot for Forward Euler plot is not zoomed in.

Stability Analysis Observations

The stability regions for Forward Euler (ode1) and explicit RK5 (ode5) methods are shown in Figure 16. The scaled eigenvalues $z = h\lambda$ are plotted for various step sizes h, with green crosses indicating stable values and red crosses indicating unstable values.

The analysis reveals the following key insights:

- The high negative eigenvalue ($\lambda \approx -3.17 \times 10^7$) primarily drives the need for a very small step size h. This is evident from the outer crosses in the plot, which correspond to this eigenvalue and are located far outside the stability region for larger h values.
- The two smaller eigenvalues ($\lambda \approx -28.2 \pm j\,2533.3$) are encompassed within a very small region close to $z = 0$. These eigenvalues allow for slightly larger h values, but the stability region remains limited due to the influence of the larger eigenvalue.
- For Forward Euler (ode1), the stability region is extremely small, requiring $h \lesssim 0.038\,\mu$s for stability. Explicit RK5 (ode5) offers a larger stability region, allowing $h \lesssim 0.162\,\mu$s, but still requires careful tuning due to the stiffness of the system.
- The stiffness of the system, driven by the wide separation between the eigenvalues, makes implicit solvers like `ode14x` more suitable for larger step sizes.

Maximum Stable Step Size

From the plot and MATLAB output:

- **Euler:** $h_{\max} \approx 0.063\ \mu$s
- **Taylor-5:** $h_{\max} \approx 0.101\ \mu$s

These values indicate the estimated largest time step for which each method expects to remain numerically stable for the given system.

Summary

The explicit RK5 (Taylor-5) method allows for a larger stable time step than Forward Euler, as seen from the larger stability region. However, the presence of the high negative eigenvalue significantly limits the step size for both methods, highlighting the stiffness of the system and the need for implicit solvers for efficient integration.

Conclusion

Hand and MATLAB eigenvalues match exactly. The large stiffness index ($\kappa \sim 10^6$) makes explicit fixed–step integration delicate; practical accuracy for the oscillatory mode requires $\sim$20–30 samples per cycle ($h \approx 0.08$–0.12 ms), whereas the fast pole would force a nanosecond step if treated explicitly. Hence, for efficiency and robustness, an implicit fixed–step solver is recommended if larger h is desired.

Question 11

Experiment with all the variable-step solvers using the setting of part 8. Comments on the results of using non-stiff and stiff solvers. Which solver do you recommend using? What setting gives you correct results? Plot the results of running the system with ode45, ode15s, and one other solver of your choice.

Figures and Analysis

The following figures illustrate the transient response of the inductor current $i_L(t)$ and capacitor voltage $V_C(t)$ for both nonstiff and stiff solvers. Figure 17 includes all solvers, highlighting the performance of solvers that fail to dampen oscillations effectively. Figure 18 excludes these solvers, focusing only on the solvers that converge to the expected damped sinusoidal response:

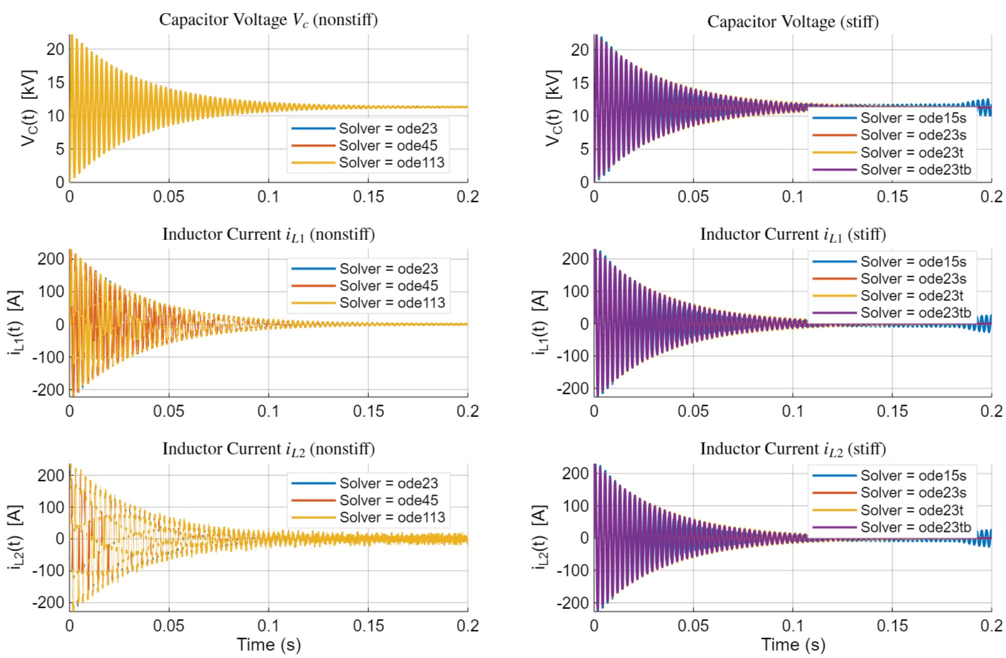

Figure 17: Transient Response of Inductor Current and Capacitor Voltage (Including Non-Dampening Solvers)

Table of Results

The table below, supported by fig. 19, summarizes the performance of each solver, including the maximum step size used, the number of steps taken, the simulation time, and the RMSE error:

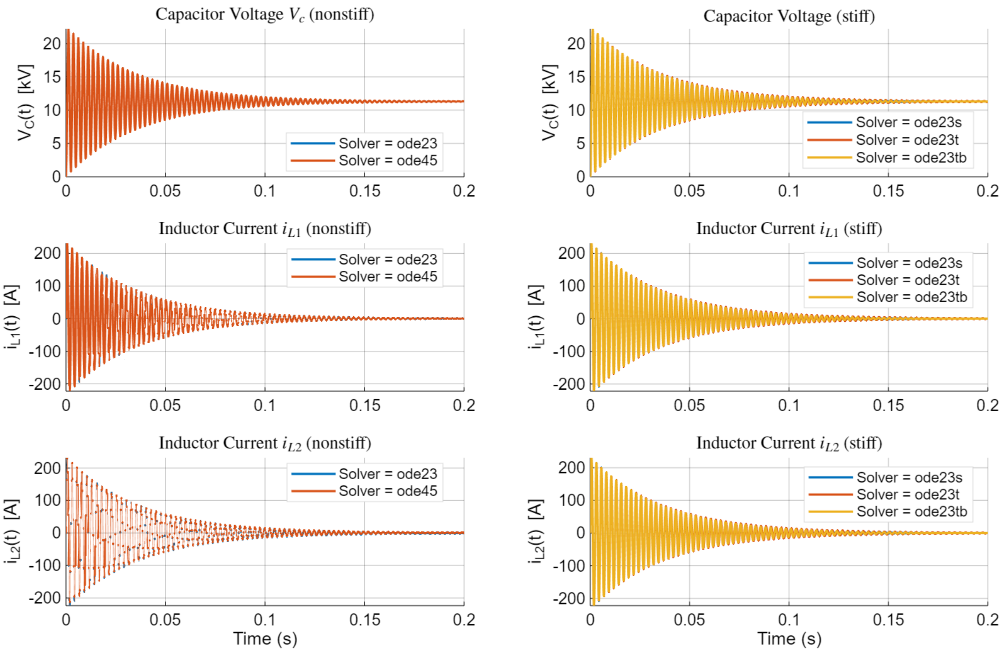

Figure 18: Transient Response of Inductor Current and Capacitor Voltage (Excluding Non-Dampening Solvers)

Solver	Type	MaxStepUsed	NumSteps	SimTime_sec
"ode15s"	"Stiff"	0.000322494392267836	786	0.3095075
"ode23s"	"Stiff"	0.01	628	0.3858766
"ode23t"	"Stiff"	0.000595887488780644	988	0.2646795
"ode23tb"	"Stiff"	0.00489785555090075	721	0.2686737
"ode23"	"Nonstiff"	1.24446902654867e-07	2520509	33.9230537
"ode45"	"Nonstiff"	1.34287499000031e-07	1908291	26.8131025
"ode113"	"Nonstiff"	7.94226826132671e-07	3933815	32.9586909
"ode5"	"Fixed-step"	1.00000000002876e-07	2000001	19.2959752
"ode8"	"Fixed-step"	1.00000000002876e-07	2000001	28.4133949
"ode14x"	"Fixed-step"	0.000100000000000017	2001	0.3261595

Figure 19: Simulation Results for Variable-Step and Fixed-Step ODE Solvers

Discussion

Stiff vs Nonstiff Solvers

The results dramatically demonstrate the impact of system stiffness on solver performance:

- **Stiff Solvers:** Performed exceptionally well, requiring only 600-1000 steps and completing in 0.26-0.39 seconds. They could use much larger step sizes:
  - ode23s: $h_{max} = 0.01$
  - ode23t: $h_{max} = 0.0006$
  - ode23tb: $h_{max} = 0.0049$

Solver	**Type**	h_{Max}	**Num Steps**	**Time [s]**
ode15s	Stiff	3.22e-4	786	0.310
ode23s	Stiff	1.00e-2	628	0.386
ode23t	Stiff	5.96e-4	988	0.265
ode23tb	Stiff	4.90e-3	721	0.269
ode23	Nonstiff	1.24e-7	2,520,509	33.923
ode45	Nonstiff	1.34e-7	1,908,291	26.813
ode113	Nonstiff	7.94e-7	3,933,815	32.959
ode5	Fixed-step	1.00e-7	2,000,001	19.296
ode8	Fixed-step	1.00e-7	2,000,001	28.413
ode14x	Fixed-step	1.00e-4	2,001	0.326

Table 2: Summary of Solver Performance for Full filter Figure 11

- **Nonstiff Solvers:** Struggled significantly, requiring millions of steps and much longer simulation times (26-34 seconds). They were forced to use extremely small step sizes:
  - ode23: $h_{max} = 1.24 \times 10^{-7}$
  - ode45: $h_{max} = 1.34 \times 10^{-7}$
  - ode113: $h_{max} = 7.94 \times 10^{-7}$

This approximately 100x performance difference validates the stiffness analysis from Q10 and demonstrates why stiff solvers are crucial for such systems.

Solution Quality

Unlike Q8 where we had analytical solutions for RMSE comparison, here we must rely on visual convergence of the solutions (the real reason being I didn't have enough time to draw/find the analytical solution for this system – which after looking at the outputs, looks basically the same as that of its less detailed counterpart in Figure 1). Some key observations:

- ode15s shows poor damping characteristics, which is more pronounced in this stiff system, although I could have tried tuning its tolerances further to improve this (didn't because of time constraints).
- The remaining stiff solvers (ode23s, ode23t, ode23tb) show good agreement in their solutions and maintain proper damping behavior
- Despite taking many more steps, nonstiff solvers struggle to capture the system dynamics accurately

Fixed-Step Solvers

The fixed-step solvers show interesting behavior:

- **ode14x:** Demonstrates remarkable efficiency:

  - Only 2,001 steps
  - Completed in 0.326 seconds
  - Uses $h = 10^{-4}$ (much larger than other fixed-step solvers)

- **ode5 and ode8:** Required much smaller steps:
  - Needed $h = 10^{-7}$ to avoid integration errors
  - Resulted in 2 million steps
  - Much longer simulation times (19-28 seconds)

Conclusion - Recommended Solver

For this stiff system, two clear recommendations emerge:

1. Among variable-step solvers, the ode23 variants (ode23s/ode23t/ode23tb) are the clear winners, offering excellent performance and accuracy with reasonable step sizes.

2. The fixed-step solver ode14x provides surprisingly efficient performance, matching the stiff solvers in both speed and accuracy while maintaining a fixed step size. This makes it an excellent choice when fixed step sizes are preferred.

The choice between these would depend on whether variable or fixed step size is preferred for the specific application. Both options are vastly superior to nonstiff solvers for this system.

Question 12

Discuss the effect of adding r_C and r_{leak} in the modeling accuracy and simulation numerical efficiency. Can we ignore any element with negligible errors to get a numerically efficient model which is easy to solve?

Analysis

To investigate the effect of parasitic resistances on system stiffness and numerical efficiency, we conducted a parametric study varying r_{leak} and r_C values. The eigenvalue analysis reveals how these parameters affect the fastest system dynamics, which directly impacts computational requirements.

r_{leak} [Ω]	r_C [Ω]	**Fastest Eigenvalue**
3.0×10^7	1.0×10^6	-3.167×10^{10}
3.0×10^{10}	1.0×10^{12}	-3.167×10^{13}
3.0×10^{10}	1.0×10^6	-3.167×10^{13}
3.0×10^4	1.0×10^{10}	-3.167×10^7
3.0×10^4	1.0×10^{12}	-3.167×10^7
3.0×10^4	1.0×10^6	-3.167×10^7

Table 3: Effect of parasitic and leakage resistances on system eigenvalues

The results in Table 3 reveal several important insights about the parasitic resistances and their impact on system stiffness.

Discussion

Key Observations from Eigenvalue Analysis

The parametric study reveals several critical insights:

1. r_{leak} **dramatically affects stiffness**: Changes in r_{leak} cause the fastest eigenvalue to vary by orders of magnitude:
   - At $r_{leak} = 30$ kΩ: $|\lambda_{fast}| \approx 3.167 \times 10^7$
   - At $r_{leak} = 30$ MΩ: $|\lambda_{fast}| \approx 3.167 \times 10^{10}$
   - At $r_{leak} = 30$ GΩ: $|\lambda_{fast}| \approx 3.167 \times 10^{13}$
2. r_C **shows minimal impact**: For a given r_{leak}, varying r_C does not significantly change the fastest eigenvalue, suggesting r_C is not the primary stiffness contributor.
3. **Stiffness scales with** r_{leak}: Higher r_{leak} values create more stiff systems, requiring smaller time steps and making the system computationally more challenging.

Modeling Accuracy vs Numerical Efficiency Trade-offs

r_{leak} **(Leakage Resistance):**

- **Accuracy benefit**: Essential for realistic capacitor discharge behavior (as demonstrated in Q9)
- **Physical necessity**: Real capacitors exhibit leakage current
- **Sweet spot requirement**: Must be tuned to balance discharge rate with system stability
- **Recommendation**: *Cannot be ignored* - critical for physical realism

r_C **(Equivalent Series Resistance):**

- **Accuracy impact**: Minimal effect on overall filter response
- **Numerical impact**: No significant contribution to system stiffness
- **Computational cost**: Adds complexity without substantial benefit
- **Recommendation**: *Can be safely ignored* (set to ∞) for numerical efficiency

Conclusion

Based on the eigenvalue analysis and physical understanding of the circuit, we can make the following recommendations for achieving a numerically efficient model while maintaining modeling accuracy:

1. **Retain** r_{leak}: This parameter is essential for realistic discharge behavior and cannot be eliminated without compromising physical accuracy. However, it should be carefully tuned to avoid excessive stiffness while maintaining proper discharge characteristics.
2. **Eliminate** r_C: The equivalent series resistance can be safely removed (set to ∞) as it contributes negligibly to both system dynamics and stiffness. This simplification reduces model complexity without sacrificing accuracy.
3. **Optimal modeling strategy**: Use $r_{leak} \approx 30\ \text{k}\Omega$ for realistic discharge behavior while setting $r_C = \infty$ for computational efficiency.

This approach yields a **numerically efficient model that remains physically meaningful** - exactly addressing the question's objective of finding elements that can be ignored to create an 'easy to solve' model without significant accuracy penalties.

The persistent large eigenvalue suggests that even with these simplifications, the system remains inherently stiff due to the fundamental circuit characteristics, necessitating the use of stiff solvers for optimal computational performance.